

STUDENT PERFORMANCE ANALYSIS PROJECT

1. Title Page

Student Performance Analysis
Data Analysis Project

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Project Type: Data Science & Data Analytics

Tools Used: Python, Pandas, Matplotlib, Jupyter Notebook

2. Project Overview

The Student Performance Analysis project focuses on analyzing academic and personal factors that may be associated with students final academic performance.

The dataset contains information about **5,000 students** and includes variables related to students' academic scores, attendance, study habits, previous academic performance, gender, parental education, internet access, and extracurricular activities.

The main objective of this project is to clean and analyse the dataset, calculate important performance indicators, identify patterns in academic performance, and understand relationships between different student-related factors and final percentage.

The analysis was performed using Python-based data analysis and visualization techniques.

3. Dataset Overview

The dataset contains records for **5,000 students**.

The major categories of information analysed include:

- Student academic scores
- Final percentage
- Attendance percentage
- Study hours per day
- Previous year score
- Gender
- Parental education
- Internet access

- Extracurricular activities
- Performance level
- Pass/Fail status

The dataset provides a useful basis for understanding student performance and comparing different academic and demographic factors.

4. Data Cleaning & Preprocessing

Before performing the analysis, the dataset was prepared for analysis using Python and Pandas.

The preprocessing process included:

- Loading the dataset into the Python environment.
- Examining the dataset structure and columns.
- Checking data types.
- Checking for missing values.
- Identifying and handling data-quality issues where required.
- Converting columns into appropriate data types.
- Checking numerical variables for consistency.
- Preparing categorical variables for analysis.
- Verifying the final dataset before performing analysis.

These preprocessing steps helped ensure that the data was suitable for statistical analysis and visualization.

5. Feature Engineering

Additional analytical features were created to make the dataset easier to analyse.

The main derived features included:

Final Percentage

A final percentage measure was used to represent the overall academic performance of students.

Performance Level

Students were categorized into different performance levels based on their final performance.

The resulting categories were:

- **Excellent**
- **Good**
- **Average**
- **Poor**

Pass/Fail Status

Students were also classified according to their final academic result to calculate the overall pass rate.

These engineered features allowed the analysis to compare students across different performance groups.

6. Key Performance Indicators

The major KPIs obtained from the analysis are:

KPI	Result
Total Students	5,000
Average Final Percentage	67.48%
Average Attendance	74.92%
Average Study Hours	3.29 hours/day
Overall Pass Rate	94.70%

These KPIs provide an overall summary of the academic performance and study-related characteristics of the students.

The overall pass rate of **94.70%** indicates that most students in the dataset successfully passed.

7. Academic Performance Analysis

Academic performance was analysed using subject scores and final percentage.

The analysis showed differences between the average scores of the subjects.

Subject Performance

- **English has the highest average subject score.**
- **Science has the lowest average subject score.**

This indicates that English was the strongest subject based on the average score, while Science had comparatively lower average performance.

Subject-wise analysis can help identify areas where students are performing well and subjects where additional academic support may be useful.

The analysis also showed that English Score has a relatively strong relationship with Final Percentage, with a correlation of **0.592**.

8. Attendance Analysis

Attendance was analysed to understand its relationship with students' final academic performance.

The average attendance of students was:

74.92%

However, the correlation between Attendance and Final Percentage was approximately:

0.001

This is a very weak relationship in this dataset.

Therefore, based on this analysis alone, attendance does not show a meaningful linear relationship with final percentage.

It is important to note that correlation describes an association and does not prove that one factor causes another.

9. Study Hours Analysis

Study hours were analysed as one of the factors related to student performance.

The average study time was:

3.29 hours per day

The analysis examined study hours in relation to final percentage to understand whether students who reported different amounts of study time showed different levels of academic performance.

The results from the analysis should be interpreted together with other academic and student-related factors rather than considering study hours alone.

10. Gender Analysis

Gender-wise analysis was performed to compare academic performance between different gender groups.

The analysis allows the distribution and average academic performance of students to be compared across gender categories.

Such comparisons can help identify whether noticeable differences exist between groups in the dataset.

However, the analysis should be interpreted as a description of this particular dataset and should not be generalized to all students.

11. Previous Year Performance

Previous Year Score was analysed to determine whether previous academic performance was associated with Final Percentage.

The correlation between Previous Year Score and Final Percentage was:

-0.017

This represents a very weak negative linear relationship.

Therefore, in this dataset, Previous Year Score did not show a meaningful linear relationship with Final Percentage.

This result is important because it shows that not every academic variable necessarily has a strong relationship with the final outcome in this particular dataset.

12. Parental Education & Internet Access

The analysis also considered factors such as:

- Parental education
- Internet access

These variables were compared with student academic performance to understand whether differences could be observed across the respective categories.

Parental education and internet access can provide useful context when studying student performance because they represent aspects of students' learning environment and access to educational resources.

The analysis focuses on associations observed in the dataset and does not establish causal relationships.

13. Extracurricular Activities

Extracurricular activity status was also included in the analysis.

Students were compared based on their participation in extracurricular activities to examine possible differences in academic performance.

This analysis provides an additional perspective on the relationship between academic performance and activities outside regular academic study.

The results should be interpreted as dataset-specific observations rather than general conclusions about extracurricular activities and academic success.

14. Correlation Analysis

Correlation analysis was used to examine the linear relationship between selected numerical variables and Final Percentage.

The important correlation results were:

Variable	Correlation with Final Percentage
English Score	0.592
Attendance	0.001
Previous Year Score	-0.017

Interpretation

English Score (0.592)

English Score has the strongest relationship with Final Percentage among the variables listed above. The positive correlation indicates that higher English scores tend to be associated with higher final percentages in this dataset.

Attendance (0.001)

Attendance has an almost zero correlation with Final Percentage, indicating a very weak linear relationship.

Previous Year Score (-0.017)

Previous Year Score has a very weak negative correlation with Final Percentage and therefore does not show a meaningful linear relationship in this dataset.

Correlation should not be interpreted as proof of causation.

15. Key Findings

The major findings from the Student Performance Analysis are:

1. The dataset contains **5,000 students**.
 2. The average Final Percentage is **67.48%**.
 3. The average Attendance is **74.92%**.
 4. Students study for an average of **3.29 hours per day**.
 5. The overall Pass Rate is **94.70%**.
 6. **English has the highest average subject score.**
 7. **Science has the lowest average subject score.**
 8. English Score has the strongest reported relationship with Final Percentage, with a correlation of **0.592**.
 9. Attendance has an extremely weak correlation with Final Percentage (**0.001**).
 10. Previous Year Score has a very weak negative correlation with Final Percentage (**-0.017**).
 11. Most students belong to the **Good** and **Average** performance categories.
 12. Performance-level distribution consists of:
 - Good: **2,191 students**
 - Average: **1,951 students**
 - Excellent: **593 students**
 - Poor: **265 students**
 13. Performance-level analysis provides a clear way to identify high-, medium-, and low-performing students.
 14. Subject-wise analysis helps identify stronger and weaker academic areas.
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16. Recommendations

Based on the findings from the analysis, the following recommendations can be considered:

1. Focus on weaker subjects

Since Science has the lowest average subject score, additional academic support and targeted practice could be considered for Science.

2. Strengthen English performance

English has the highest average subject score and also shows the strongest reported correlation with Final Percentage. Maintaining strong English performance may be useful when supporting overall academic achievement.

3. Support average and poor performers

The performance-level analysis can be used to identify students who may benefit from additional academic support, particularly students in the Average and Poor categories.

4. Use data-driven academic monitoring

Teachers and institutions can regularly monitor subject scores, final percentages, attendance, and other available student information to identify areas requiring attention.

5. Avoid relying on a single factor

The analysis shows that some variables have very weak correlations with Final Percentage. Therefore, student performance should be evaluated using multiple factors rather than relying on one variable alone.

6. Use correlation carefully

Correlation results should be used to identify relationships for further investigation. They should not automatically be interpreted as evidence that one variable causes changes in another.

17. Conclusion

The Student Performance Analysis project provided an overview of the academic performance of **5,000 students** using Python-based data analysis techniques.

The students had an average Final Percentage of **67.48%**, average Attendance of **74.92%**, and average Study Hours of **3.29 hours per day**. The overall Pass Rate was **94.70%**.

The subject-wise analysis showed that **English had the highest average score**, while **Science had the lowest average score**.

The correlation analysis produced an important finding: **English Score had the strongest reported relationship with Final Percentage, with a correlation of 0.592**. In comparison, Attendance (**0.001**) and Previous Year Score (**-0.017**) showed very weak relationships with Final Percentage.

The performance-level analysis showed that **Good** was the largest performance category with **2,191 students**, followed by **Average with 1,951 students**, **Excellent with 593 students**, and **Poor with 265 students**.

Overall, this project demonstrates how data analysis can be used to summarize student performance, identify academic patterns, compare different groups, and generate meaningful insights from educational data.

18. Tools Used

The following tools and technologies were used for this project:

Python

Used as the primary programming language for data analysis.

Pandas

Used for:

- Data loading
- Data cleaning
- Data manipulation
- Grouping
- Aggregation
- Feature creation
- Statistical analysis

NumPy

Used for numerical operations and data processing.

Matplotlib

Used to create data visualizations and charts.

Jupyter Notebook

Used as the development environment for writing, executing, and documenting the analysis.

PROJECT SUMMARY

Project: Student Performance Analysis

Dataset Size: 5,000 students

Average Final Percentage: 67.48%

Average Attendance: 74.92%

Average Study Hours: 3.29 hours/day

Overall Pass Rate: 94.70%

Highest Average Subject: English

Lowest Average Subject: Science

Strongest Reported Correlation: English Score → Final Percentage (**0.592**)

Performance Categories:

- Good: 2,191
- Average: 1,951
- Excellent: 593
- Poor: 265